

STATISTICS

Identify the research problem

Preliminary literature review to clarify research problem, draft hypothesis/research question

Draft precise statement of the problem

Identify purpose of the study, information needs, conceptual & operational measures, use of information once study is complete

Conduct a comprehensive literature review

Critically review previous studies, identify additional information needed before going further; identify how will approach the issue

Revise hypothesis(es) & state in measurable terms

Specify independent & dependent variables, establish their relationships, state questions to answer
Develop valid & reliable measures for dependent variable(s)

Identify the population—census, sample; if sample, determine size & sampling design

Identify population, determine if will conduct census or sample, if a sample, determine size; define sampling design

Choose appropriate test statistics & statistical analysis design

Identify research design

Specify how existing data will be used, identify what will be measured or observed; choose research methods & techniques; identify who will collect data & how long it will take

Collect data

Analyze data/draw conclusions

Analyze data according to data analysis design, address hypothesis/research questions—has hypothesis been proved or disproved? Is our information valid & reliable?

DEFINITION	TRANSFORMATION	DESCRIPTION	PROCEDURE
Portrays modal pattern & range of variation	Frequency Distribution	Decide on types of variable transformations; describe profiles & variation population	Raw scores ranked from low to high; grouped scores; frequency of appearances of scores
Defines average of a set of scores	Mean	Describes the predominant trend at the interval or ratio level	Add scores & divide by number of scores
Defines the midpoint in a range of scores	Median	Describes the predominant trend at the ordinal, interval, & ratio levels	Rank scores; find the midpoint; divide a distribution of scores exactly in half
Defines the most frequent score	Mode	Describes the predominant trend at the nominal, ordinal, interval, & ratio levels	Scores are ranked & frequencies established; the modal score or group score has the highest frequency
Difference between highest & lowest scores for a variable	Range	Describes the spread of scores to show the extent of variability in the population	Highest & lowest scores
Calculates the average of the summed squares deviation of each score from the mean	Variance	Used instead of standard deviation in inferential statistical analysis	Standard deviation squared
A measure of the spread of scores around the mean	Standard Deviation	Summarizes variability in a population; used for descriptive purposes instead of the range	--
Compares mean for two unrelated samples	t-test	Used to determine whether there are statistically significant differences in the means of 2 independent or 2 related populations	Compares means
Determines whether the relationship between two nominal variables is greater than chance	Chi-square	Explanation of the relation between variables such that one is logically seen as causing the 2 nd ; tests covariation between 2 nominal variables	Bivariate (2x2) table with independent variable at the top & dependent variable along the side
Determines whether the relationship between two nominal variables is greater than chance	Fisher's Exact Test	For use with 2x2 tables where frequencies for any cell in a table are less than 5; used most often with small samples; does not suggest direction of relationship	Identical to chi-square
Measures the association between two ordinal variables	Gamma	Tells whether knowing the ranking of pair of people on one variable enhances capacity to predict ranking on a second	--
Determines whether there is a statistically significant relationship between two interval or ratio measures	Pearson's product moment correlation (r)	Can be used with interval/interval variables; interval & ordinal, & interval nominal variables	3x3 table; formula is the same ranked pairs - opposite ranked pairs/same ranked pairs + opposite ranked pairs
The relationship between groupings of means in a population & a continuous dependent variable	Eta (r)	Tests covariation between nominal-interval & ordinal-interval variables; addresses nonlinear relations with interval variables	Based on guessing the value of one variable by knowing the other
Tests the hypothesis that the means of two or more populations on a single variable are not equal to one another	ANOVA	Inferential: compares group means in two or more groups within the same population or between two different populations at two points in time	(Sub) Groups mean based on researcher opinion; relates means to outcome or dependent variable; arrives at probability score
Assesses associations among continuous variables to determine statistically defined variable groupings	Factor analysis	Reduces variables to scales consisting of similar continuous variables; factors allowed to emerge from the data	Compares means of two or more independent populations on a single variable
Predictive model with at least one categorical independent variable & one continuous dependent variable	One-way ANOVA	Inferential: used to determine relationships between categorical & continuous variables	Requires a predictive model with at least one categorical independent variable & one continuous dependent variable
Tests the effect of two categorical independent variables on a continuous dependent variable	Two-way ANOVA	Inferential: used to calculate relationships between categorical & continuous variables	Predictive model with at least two independent variables & one continuous dependent variable
Tests the effects of more than two categorical independent variables on a continuous dependent variable	Multiple-way ANOVA	Permits an examination of the effects of multiple independent categorical variables on a continuous dependent variable	Requires a predictive model with more than two categorical independent variables & one continuous variable
Association between two interval or ratio variables: y is a function of X	Linear Regression	Regression equation representing the geometric line that comes closest to the distribution of points on the x-y axis with values of x, we can predict y	Y causes x so the value of y determines the value of x
Association between multiple interval or ratio independent variables & a dependent variable	Multiple Regression	When a dependent variable is affected simultaneously by several independent variables	Multiple independent variables account for more of the variance than does a single independent variable
Assumes that the values of one variable are caused by the values of another	Path analysis	Considers patterns of relations among variables & calculates the strength of the relationships in a multivariate model	Must identify independent & dependent variables